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PROGRAMMING IN JAVA

Assignment 1

TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 10 × 1 = 10**

QUESTION 1:

Which of the following is NOT a primitive data type in Java??

a. int
b. boolean
c. String
d. char

Correct Answer: c

Detailed Solution:

String is not a primitive data type.

QUESTION 2:

Consider the following program.

```
public class Question{
 public static void main(String[] args){
 byte x = 127;
 x++;
 ++x;
 System.out.print(x);
 }
}
```

What is the output of the above code?

a. 127
b. -127

c. 129
d. 2

Correct Answer: b

Detailed Solution:

Range of byte data in java is -128 to 127. But the byte data type in java is cyclic in nature.

QUESTION 3:

Which of the following concept that Java doesn't support?

a. inheritance
b. encapsulation
c. pointer
d. array

Correct Answer: c

Detailed Solution:

Java does not support pointer.

QUESTION 4:

Which of the following is not a keyword in java?

a. import
b. super
c. method
d. class

Correct Answer: c

Detailed Solution:
Here, import, super and class are reserved keywords in Java, which cannot be used for naming a variable or class.

QUESTION 5:

Consider the following program.

```
public class Question{
 public static void main(String[] args){
int[] x = {12, 20, 015};
 for(int i = 0; i < x.length; i++){
System.out.print(x[i] + " ");  }
 }
}
```

What will be the output of the program if it is executed?

a. 12 20 13
b. 12 20 15
c. 12 20 015
d. 12 20 F

Correct Answer: a

Detailed Solution:

015 is an octal number, its equivalent decimal number is 13.

QUESTION 6:

Match the following java terms with their descriptions?

A. JDK	1. A software development kit that provides all the necessary tools and libraries to develop, compile, and run Java applications.
B. JRE	2. A software package that provides the necessary runtime environment for executing Java applications.

C. JVM	3. An abstract machine that executes java bytecode.

Choose the correct option.

a. A-1 , B-2, C-3
b. A-2 , B-3, C-1
c. A-3, B-2 ,C-1
d. A-3, B-1,C-2

Correct Answer: a

Detailed Solution:

Option a is correct.

QUESTION 7:

Consider the following program.

```
public class Question{
 public static void main(String[] args){
short x = 20;
 x = x * 2;
 System.out.print(x);  }
 }
```

What will be the output of the program if it is executed ?

a. 20
b. 2
c. Compiler error
d. 40

Correct Answer: c

Detailed Solution:

This will give compile error - “Lossy conversion from int to short”.

QUESTION 8:

What is the output of the following code ?

```
public class Question{
 public static void main(String[] args){
 boolean x = true;
 boolean y = false;
 boolean z=(x==y);
 System.out.println(z);
 }
}
```

a. null
b. true
c. false
d. 1

Correct Answer: c

Detailed Solution:
Test by run.

QUESTION 9:

Which program is used to compile Java source code into bytecode?

a. javap
b. javac
c. java
d. javad

Correct Answer: b

Detailed Solution:

The JDK includes a tool, 'javac' that compiles from Java source code to a target of Java bytecodes. The program 'java' is used to execute Java bytecode. Note that there is no program like 'javap' and 'javad'.

QUESTION 10:

Consider the following program.

```
public class Question{
 public static void main(String[] args){
 int x = 5;
 x *= (2 + 8);
 System.out.println(x);
 }
}
```

What will be the output of the program if it is executed ?

a. 50
b. 10
c. Compiler error
d. 5

Correct Answer: a

Detailed Solution:

Here, x* = 2 + 8 is equivalent to x * (2+ 8) = x * 10. Therefore, x = 50.

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PROGRAMMING IN JAVA

Assignment 2

TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 10 × 1 = 10**

QUESTION 1:

Consider the following code segment:

```
1 class Question{
2 public static void main(String args){
3 System.out.print("Welcome to NPTEL");
4 }
5 }
```

Identify the line number(s) where there is/are error(s) in the above code.

a. 1
b. 2
c. 3
d. 4 and 5

Correct Answer: b.

Detailed Solution: The String argument in the main method is an array hence the args should be changed to args[].

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QUESTION 2:

Consider the following code segment:

```
1 class Question{
2 public static void main(String[] param){
3 System.out.print("Welcome to NPTEL");}}
4 }
5 }
```

Identify the line number(s) where there is error in the above code.

a. 1
b. 2
c. 3
d. 4 and 5

Correct Answer: d.

Detailed Solution: The String argument in the main method can have any valid variable name, so line 2 have no error. The closing brackets in line 4 and 5 are extra as the scope is already ended in line 3.

QUESTION 3:

Consider the following code segment:

```
1 class Question{
2 public static void main(String args[]){
3 for(int n=1; n<=10; n++){
4 System.out.print(n+ " ");
5 }
6 }
7 }
```

Select the correct output description for the above code.

a. Prints first n natural numbers in a single line.
b. Prints first n natural numbers, one number in a single line.
c. Prints first 10 natural numbers in a single line.
d. Prints first 10 natural numbers in a single line with no spaces.

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Correct Answer: c.

Detailed Solution: The code segment prints first 10 natural numbers in a single line with spaces in between.

QUESTION 4:

Consider the following code segment:

```
1 class Question{
2 public static void main(String args[]){
3 for(int n=1; n<=100; n++){
4 System.out.print(n+ " ");
5 }
6 }
7 }
```

Modify the above code such that it prints all the even numbers till 100.

a. Replace line 3 with for(int n=2; n<=100; n++){
b. Replace line 3 with for(int n=1; n<=100; n+=2){
c. Replace line 4 with System.out.print(++n + " ");
d. Both option a and c.

Correct Answer: c.

Detailed Solution: If option a is considered, then all natural numbers starting from 2 will be printed. Considering option b, instead of even numbers, all odd numbers will be printed. Option c will print all the even numbers, as pre-increment operator in the print statement increments one value and variable updating parameter post-increments one value. Option d, again prints all the odd numbers instead of the even numbers.

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QUESTION 5:

Following is a program given for this question.

```
public class Question{
      public static void main(String[] args){
 int[] x = {222, 220, 022};
```

```
 for(int i = 0; i < x.length; i++){
 System.out.print(x[i] + "");
 }
 }
}
```

What will be the output of the above program?

a. 22221018
b. 22222018
c. 22101018
d. 22221218

Correct Answer: b

Detailed Solution:
Since, `int[2]` numeral value have 0 in leading, therefore, it's taken as octal base (8) with the output being it's corresponding decimal value.

QUESTION 6:

Consider the following incorrect program.

```
1 public class Question{
2 public static void main(String[] args){
3 short x = 10;
4 x = x * 5;
5 System.out.print(x);
7 }
8 }
```

How to correct the above code segment?

a. Change line 4 as `x =(short) (x * 5);`
b. Change line 4 as `x =(short) x * 5;`
c. Change line 4 as `x =(short) x * (short) 5;`
d. Change line 4 as `(short) x = x * 5;`

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Correct Answer: a

Detailed Solution:
It works if the proper type casting is followed, for example, `x = (short) ( x * 5);`

QUESTION 7:

What is the output of the following program?

```
public class Question{
 public static void main(String[] args){
 int[] A = {1,0,1};
 for(int i = 0; i < A.length; i++){
 A[i] = A[(A[i] + 3) % A.length];
 }
 for(int i = 0; i < A.length; i++){
 System.out.print(A[i]);
 }
 }
}
```

a. 210
b. 120
c. 012
d. 000

Correct Answer: d

Detailed Solution:
The output can be checked by execution.

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QUESTION 8:

Consider the following piece of code.

```
public class Question{
 public static void main(String[] args){
 String str = "anpctdelqjpava";
 System.out.println(str.substring(1,3)+str.substring(4,5)+
str.substring(6,8));
 }
}
```

Which of the following option is the output of the above program?

a. java
b. npteljava
c. nptel java
d. nptel

Correct Answer: d

Detailed Solution:
The output can be checked by execution.

QUESTION 9:

What is the output of the following program?

```
public class Main{
      public void static main(String args[]){
            char a = 'a';
            int b = 20;
            System.out.println(a+b);
      }
}
```

a. 60
b. 117
c. 33
d. Compilation error

Correct Answer: d

Detailed Solution:

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There is a compilation error due to incorrect syntax in the statement `public void static main(String args[]);` it should be `public static void main(String args[]){`.

QUESTION 10:

Consider the following program.

```
public class Question{
 public static void main(String[] args){
 int x = 5;
 x *= (2 + 8);
 System.out.println(x);
 }
}
```

What will be the output of the program if it is executed?

a. 50
b. 10
c. Compiler error
d. 5

Correct Answer: a

Detailed Solution:

Here, x* = 2 + 8 is equivalent to x * (2+ 8) = x * 10. Therefore, x = 50.

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PROGRAMMING IN JAVA

Assignment 3

TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 10 × 1 = 10**

QUESTION 1:

In which of the following scenario(s), the static block is used in Java?

a. To create static variables.
b. To initialize instance variables.
c. To initialize static variables.

d. To create new objects.

Correct Answer: c

Detailed Solution:

The static keyword can be used to create a block to be used to initialize static variables. This static block executes when classloader loads the class.

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QUESTION 2 :

Consider the following piece of code.

```
public class Question {
 private static int count = 0;

 public static void main(String[] args) {
 incrementCount();
 System.out.println("Count: " + count);
 }

 _______________ incrementCount() {
 count++;
 }
}
```

Fill in the blank with the appropriate keyword(s) from the list given below so that the program compiles successfully.

a. public void
b. private void
c. public static void
d. private static void

Correct Answer: c, d

Detailed Solution:

Option c and d are correct keyword(s).

QUESTION 3:

Consider the following piece of code.

```
class A {
 public void display() {
 System.out.println("A's display method");  }
}

class B extends A {
 public void display() {
 System.out.println("B's display method");  }
}

public class Main {
 public static void main(String[] args) {
 A a = new B();
 a.display();
 ((B) a).display();
 }
}
```

What is the output of the above code?

a. A's display method
   B's display method
b. A's display method
   A's display method
c. B's display method
   B's display method
d. B's display method
   A's display method

Correct Answer: c

Detailed Solution:
Test by run.

QUESTION 4:

Which of the following statement(s) is/are false?

a. You can write a new instance method in the subclass with the same signature as the one in the superclass, thus overriding it.
b. You can write a new static method in the subclass with the same signature as the one in the superclass, thus hiding it.
c. A subclass inherits all of its parent's public and protected members, no matter what package the subclass is in.
d. You cannot declare new methods in the subclass that are not in the superclass.

Correct Answer: d

Detailed Solution:
You can declare new methods in the subclass that are not in the superclass.

QUESTION 5:

Which of the following statement(s) is/are true?

a. You will get a compile-time error if you attempt to change an instance method in the superclass to a static method in the subclass.
b. You can prevent a class from being subclassed by using the final keyword in the class's declaration.
c. An abstract class can be instantiated.
d. Common behaviour can be defined in a superclass and inherited into a subclass using the extends keyword.

Correct Answer: a, b, d

Detailed Solution:

An abstract class cannot be instantiated.

QUESTION 6:

Consider the following program.

```
public class Question{
 public static void main(String[] args){
 String str = " programming in java ";
 System.out.println(str.substring(1,3)+str.substring(4,5)+
str.substring(6,8));
 }
}
```

What is the output of the above program?

a. prgam
b. program
c. gramm
d. ing in

Correct Answer: a

Detailed Solution:
Test by a run.

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QUESTION 7:

Consider the following piece of code.

```
class Question{
      static int a =10;
}

class Question1 extends Question{
      static int a =20;
}

public class Quest extends Question1{
      public static void main(String args[]){
            a =100;
            System.out.println(Question.a);
            System.out.println(Question1.a);
      }
}
```

Which of the following is the output of the above program?

a. 10
100

b. 10
20

c. 100
10

d. 10
10

Correct Answer: a

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Detailed Solution:
Test by run.

<u>QUESTION 8:</u>

Consider the following program.

```
class Question {
 int a=400;
 int b=200;
}
public class Child1 extends Question {
 int a=1000;
 int b=2000;
void add(int a,int b) {
 System.out.println(a+this.b-super.a);
 }
public static void main(String[] args) {
 Child1 c = new Child1();
 c.add(100,300);
 }
}
```

If the program is executed, then what will be the output from the execution? a. 1700

b. 1300
c. 0
d. 2600

Correct Answer: a

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Detailed Solution:
a = 100 , this.b=2000, super.a=400 , 100+2000-400=1700

QUESTION 9:

Which of the following statement(s) is/are true?

a. Hiding internal data from the outside world and accessing it only through publicly exposed methods is known as data encapsulation.
b. Static methods in interfaces are never inherited.
c. The term "class variable" is another name for a non-static field.
d. A local variable stores a temporary state; it is declared inside a method.

Correct Answer: a, b, d

Detailed Solution:
The term "class variable" is another name for a static field.

QUESTION 10:

All classes in java are inherited from which class?

a. java.lang.class
b. java.class.inherited
c. java.class.object
d. java.lang.Object

Correct Answer: d

Detailed Solution:

All classes in java are inherited from Object class.

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PROGRAMMING IN JAVA

Assignment 4

TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 10× 1 = 10**

QUESTION 1:

Which of the following is the correct statement for creating a package?

a. <*package name*> package;
b. package <*package name*>;
c. package;
d. <*package name*>;

Correct Answer: b

Detailed Solution:

To define a package, every source file needs to start with the statement 'package' followed by the *package name*.

QUESTION 2:

How Java Runtime Environment (JRE) knows where to look for a package that you create?

a. It searches in the current directory.
b. It searches in the location set in the CLASSPATH environment variable.
c. A user can set the path during runtime using the `-classpath` option.
d. Using the `–path` option, a user can set the path.

Correct Answer: a, b, c

Detailed Solution:

- First, by default, the Java run-time system uses the current working directory as its starting point. Thus, if your package is in a sub-directory of the current directory, it will be found.
- Second, you can specify a directory path or paths by setting the CLASSPATH environmental variable.
- Third, you can use the -classpath option with **java** and **javac** to specify the path to your classes.

QUESTION 3:

Which of the following statement(s) is/are false?

a. Java packages are hierarchical.
b. `System.out.println()` is a predefined java function.
c. Java can have a nested class structure.
d. The Java *static* keyword is a non-access modifier.

Correct Answer: a

Detailed Solution:
At first, packages appear to be hierarchical, but they are not. For example, the Java API includes a java.awt package, a java.awt.color package, a java.awt.font package, and many others that begin with java.awt. However, the java.awt.color package, the java.awt.font package, and other java.awt.xxxx packages are not included in the java.awt package. The prefix java.awt (the Java Abstract Window Toolkit) is used for a number of related packages to make the relationship evident, but not to show inclusion. All the other options are correct.

QUESTION 4:

Which of the following is/are NOT correct regarding packages in Java?

a. Java supports both pre-defined and user-defined packages.
b. Packages are used to organize a set of related classes and interfaces.
c. Pre-defined packages help to develop programs easily by providing thousands of classes.
d. Packages are used to organize only a set of related classes and not interfaces.

Correct Answer: d

Detailed Solution:

A package is a namespace that organizes a set of related classes and interfaces. It is just like a folder in your computer, where, you might keep HTML pages in one folder, images in another, and scripts or applications in yet another. Since, Java programs can be composed of hundreds or thousands of individual classes, it makes sense to keep things organized by placing related classes and interfaces into packages.

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QUESTION 5:

Consider the program given below.

```
public class Main{
      public static void main(String args[]){
            System.out.println(cos(2*PI));
      }
}
```

What will be the output if the above program is executed?

a. It will give compile-time error
b. It will give run-time error
c. 1.0
d. 3.14

Correct Answer: a

Detailed Solution:
Test by run.
The static import statement is used to import the static members (e.g., PI) of java.lang.Math. `import static java.lang.Math.*;`

QUESTION 6:

Which of the following is the minimum requirement for executing a Java program?

a. JDK
b. JRE
c. JDK without JRE
d. JRE without JDK

Correct Answer: b, d

Detailed Solution:

JRE (Java Runtime Environment) is required for the execution of the Java programs. JDK contains JRE by default. Therefore, if we talk about minimum requirement, then both **b** and **d** options are correct.

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QUESTION 7:

Which of the following is required for developing a Java program?

a. JDK
b. JRE
c. JDK without JRE
d. JRE without JDK

Correct Answer: a

Detailed Solution:

JDK (Java Development Kit) is required for developing the Java programs. Since, JDK already comes packed with JRE and there is no option to select JDK without JRE, option **a** is the only valid option**.**

QUESTION 8:

Which of the following is/are valid declaration(s) of an interface?

a.
```
public interface Question {
 void method (int value) {
 System.out.println("Nptel");
 }
 }
```

b.
```
public interface Question {
 void method (int value);
 }
```

c.
```
public interface Question {

 }
```

d.
```
public interface Question {
 default void method (int value) {
 System.out.println("Nptel");

 }
 }
```

Correct Answer: b,c,d

Detailed Solution:

In option (a), It has a method implementation in it. Only default and static methods have implementations. So, option (a) is not a valid interface. All others are valid.

QUESTION 9:

Which of the following statement(s) is/are NOT true?

a. The default package in the Java language is java.lang.
b. String is a final class and it is present in java.lang package.
c. Runnable is a class present in java.lang package.
d. Thread is a class present in java.lang package.

Correct Answer: c

Detailed Solution:

Runnable is an interface in java.lang package.

QUESTION 10:

Which of the following statement(s) is/are true?

a. With the import statement, generally import only a single package member or an entire package.
b. To import all the types contained in a particular package, use the import statement with the asterisk (*) wildcard character.
import package.*;

c. import package.A*; it used to match a subset of the classes in a package starts with "A". d. import package.A*; it generates compilation error.

Correct Answer: a,b,d

Detailed Solution:
import package.A*; it generates compilation error.

**********END**********

PROGRAMMING IN JAVA

Assignment 5

TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 10× 1 = 10**

QUESTION 1:

Consider the following program.

```
class Question
{
 int i;

 public Question(int i)
 {
 this.i = i--;
 }
}

class Question1 extends
Question {
 public Question1(int i)
 {
 super(++i);

 System.out.println(i);
 }
}

public class Check
{
 public static void main(String[] args)
{
 Question1 n = new Question1(20);  }
 }
```

If the program is executed, then what will be the output?

a. 20
b. 21
c. 19
d. 22

Correct Answer: b

Detailed Solution:
The output can be checked by execution.

QUESTION 2:

Which of the following statement(s) is/are false?

1. A class can extend more than one class.

2. An interface can extend many interfaces.

3. An interface can implement many interfaces.

4. A class can extend one class and implement many interfaces.

a. 1
b. 2
c. 3
d. 4

Correct Answer: a, c

Detailed Solution:

An interface can extend many interfaces. A class can extend one class and implement many interfaces. These statements are true.

QUESTION 3:

What is the output of the following code?

```
class Question
{
 static int x;

 static
 {
 x++;
 }

 {
 ++x;
 }
}

class Question1 extends Question
{
 static
 {
 --x;
 }

 {
 x--;
 }
}

public class Check
{
 public static void main(String[] args)
{
 System.out.println(new Question1().x);  }
}
```

a. 1
b. 2
c. 0
d. Compiler error

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Correct Answer: c

Detailed Solution:
The output can be checked by execution.

QUESTION 4:

Which of the following statement(s) is/are false?

a. Interface can provide the implementation of an abstract class.
b. The variables defined inside an interface are static and final by default.
c. An interface is used to achieve full abstraction.
d. Inside an interface, a constructor can be called using the super keyword with hierarchy.

Correct Answer: a, d

Detailed Solution:

A constructor cannot be called inside an interface using the super keyword with hierarchy.
Interface can't provide the implementation of an abstract class.

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QUESTION 5:

What is the output of the following code?

```
class Q
{
 public void disp()
 {
 System.out.println("java");
 }
}

class P extends Q
{
 public void disp()
 {
 System.out.println("nptel");   }
}

class C extends P
{
 public void disp()
```

```
 {
 super.super.disp();
 System.out.println("course");   }
}

public class Question
{
 public static void main(String[] args)
{
 C c = new C();
 c.disp();
 }
}
```

a. java
b. java
course
c. nptel
course
d. Compiler error

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Correct Answer: d

Detailed Solution:
The output can be checked by execution. In Java, it is not allowed to do super.super.

QUESTION 6:

Which of the following keywords is/are a part of exception handling? a. finally
b. throws
c. throw
d. thrown

Correct Answer: a, b, c

Detailed Solution:

Exceptional handling have 5 keywords – try, catch, throws, throw and finally.

QUESTION 7:

Consider the following piece of code.

```
interface X
{
 void display();
}

class Y implements X
{
 ___________ display()
 {
 System.out.println("Java");
 }
}

public class MainClass
{
 public static void main(String[] args)
 {
 Y r = new Y();
 r.display();
 }
}
```

Fill the blank in the above program to print the output "Java".

a. public void
b. void
c. private void
d. static void

Correct Answer: a

Detailed Solution:

Interface methods must be implemented as public. Because, interface methods are public by default and you should not reduce the visibility of any methods while overriding.

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QUESTION 8

Exception class exist in which of the following package?

a. java.void
b. java.io
c. java.lang
d. java.awt

Correct Answer: c

Detailed Solution:

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QUESTION 9:

Consider the following code .

```
interface X
{
 class Question
 {
 int i;
```

```
 public Question(int i)
 {
 this.i = ++i;
 }

 int disp()
 {
 return ++i;
 }
 }
}

public class Question1
{
 public static void main(String[] args)
 {
 __________________________ // fill in the blank
 System.out.println(c.disp());
 }
}
```

Fill in the blank with appropriate syntax for creating an object "c" to run the above program successfully.

a. X c = new X(1);
b. X c = new X( );
c. X.Question c = new X.Question(0);
d. X.Question c = new X.Question(1);

Correct Answer: c, d

Detailed Solution:
The output can be checked by execution.

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QUESTION 10:

```
class X
{
```

```
 static
 {
 Y.display();
 }
}

class Y extends X
{
 static void display()
 {
 System.out.println("Java");
 }
}

public class MainClass
{
 public static void main(String[] args)
 {
 Y.display();
 }
}
```

If the program is executed, then how many times "Java" will print?

a. 0
b. 1
c. 2
d. 3

Correct Answer: c

Detailed Solution:

The output can be checked by execution.

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**********END**********

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PROGRAMMING IN JAVA

Assignment6

TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 10 × 1 = 10**

QUESTION 1:

Which of the following is NOT a method of the Thread class in Java?

a. `isInterrupted()`
b. `interrupt()`
c. `joins()`
d. `sleep()`

Correct Answer: c

Detailed Solution:

join() is a method in the pre-defined Java class Thread but not joins(). Other methods like `isInterrupted` (),`interrupt` () and sleep() are defined in the Thread class.

QUESTION 2:

Which of the following statement(s) is/are true?

a. **`public int getId():`** returns the id of the thread.
b. **`public boolean isAlive():`** tests if the thread is alive.
c. **`public void interrupt():`** interrupts the thread.
d. **`public boolean isInterrupted():`** tests if the thread has been interrupted.

Correct Answer: a,b,c,d

Detailed Solution:

All options are correct.

QUESTION 3:

Which of the following can be used to create an instance of Thread?

a. By implementing the Runnable interface.
b. By extending the Thread class.
c. By creating a new class named Thread and calling method run ().
d. By importing the Thread class from package.

Correct Answer: a, b

Detailed Solution:

An application that creates an instance of Thread must provide the code that will run in that thread. There are two ways to do this:

- *Provide a Runnable object.* The Runnable interface defines a single method, run, meant to contain the code executed in the thread. The Runnable object is passed to the Thread constructor
- *Subclass Thread.* The Thread class itself implements Runnable, though its run method does nothing. An application can subclass Thread, providing its own implementation of run

Reference:https://docs.oracle.com/javase/tutorial/essential/concurrency/runthread.html

---

QUESTION 4:

What is the output of the following program?

```
public class Question
 {
     public static void main(String[] args) {

      try {
      int a=5/0;

 } catch (Exception e) {
```

```
catch (ArithmeticException a) {

}

}
            System.out.println("Hello World");
        }
}
```

a. Hello World
b. 5
c. Compile time error
d. ArithmeticException

Correct Answer: c

Detailed Solution:

This first handler catches exceptions of type Exception; therefore, it catches any exception, including ArithmeticException. The second handler could never be reached. This code will not compile.

QUESTION 5:

Which one of these keywords must be used to handle the exception thrown by try block in some rational manner?

a. try
b. finally
c. throw
d. catch

Correct Answer: d

Detailed Solution:
The catch block is responsible for handling the exceptions raised by try block.

QUESTION 6:

Assume the following method is properly synchronized and called from a thread A on an object B: wait(2000); After calling this method, when will the thread A become a candidate to get another turn at the CPU?

a) After thread A is notified, or after two seconds.
b) Two seconds after thread A is notified.
c) After the lock on B is released, or after two seconds.
d) Two seconds after lock B is released.

Answer : **a**

Explanation: Either of the two events (notification or wait time expiration) will make the thread become a candidate for running again

---

QUESTION 7:

The following is a simple program using the concept of thread.

```
public class Question extends Thread{
 public void run(){
 for(int i=1;i<5;i++){

 System.out.println(i++);
 }
  }
 public static void main(String args[]){
 Question t1=new Question();
 t1.run();

  }
 }
```

What is the output of the above program?

a. 1
   3
b. 1
   2

3
4

c. Runtime error

d. 1
2

Correct Answer: a

Detailed Solution: Run to see the output.

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QUESTION 8:

For the program given below, what will be the output after its execution?

```
public class Main{
public static void main(String[]args){
 Thread thread=Thread.currentThread();
 System.out.println(thread.activeCount());
 }
}
```

a. 0
b. true
c. 1
d. false

Correct Answer: c

Detailed Solution:

java.lang.Thread.activeCount() : Returns an estimate of the number of active threads in the current thread's thread group and its subgroups.

QUESTION 9:

Which of the following is a correct constructor for a thread object?

a. `Thread(Runnable a, String str);`
b. `Thread(Runnable a, int priority);`
c. `Thread(Runnable a, ThreadGroup t);`
d. `Thread(int priority);`

Correct Answer: a

Detailed Solution:
`Thread(Runnable a, String str)` creates a new Thread object. The others are not valid constructors to create a thread object.

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QUESTION 10:

Which of these keyword(s) is used to manually throw an exception?

a. try
b. finally
c. throw
d. catch

Correct Answer: c

Detailed Solution:
The throw keyword is used to manually throw an exception.

**********END**********

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PROGRAMMING IN JAVA

Assignment7

TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 10 × 1 = 10**

QUESTION 1:

Consider the following program.

```
import java.io.*;
 public class Question {
 public static void main(String[] args)
 {
 try {
 PrintWriter writer = new PrintWriter(System.out);
writer.write('4'+ '2');
 writer.close();
 }
 catch (Exception e) {
 System.out.println(e);
 }
 }
 }
```

What will be the output if the above program is executed?

a. It will give a compile-time error.
b. 102
c. 42
d. f

Correct Answer: d

Detailed Solution:

The expression `'4'+'2' add 52 with 50(i.e, Ascii value of 4 and 2)=102 and  print the corresponding char, in this case, f.`

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QUESTION 2:

Which method is used to write an array of byte to the current output stream?

a. public void write(int b)throws IOException{{
b. public void flush(byte[ ] b)throws IOException{}
c. public void write(byte[ ] b)throws IOException{}
d. public int write(int b)throws IOException{}

Correct Answer: c

Detailed Solution:

public void write(byte[ ] b)throws IOException{} is used to write an array of byte to the current output stream.

QUESTION 3:

Which of the following is NOT a Standard Stream?

a. System.in
b. System.out
c. System.err
d. System.console

Correct Answer: d

Detailed Solution:
The Java platform supports three Standard Streams: Standard Input, accessed through System.in; Standard Output, accessed through System.out; and Standard Error, accessed through System.err.

QUESTION 4:

Which of the following method(s) is not included in the OutputStream class?

a. close( )
b. write( )
c. skip( )
d. flush( )

Correct Answer: c

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Detailed Solution:

skip( ) is not a valid method in the OutputStream class.

QUESTION 5:

Which of the following methods of the BufferedReader class is used for reading lines of text from the console, the file, or other input streams?

a. read( )
b. readLine( )
c. readByte( )
d. read(byte [ ] b)

Correct Answer: b

Detailed Solution:

readLine( ) method of the BufferedReader class is used for reading lines of text from the console, the file, or other input streams.

QUESTION 6:

Consider the following program.

```
import java.io.*;
 public class files
 {
 public static void main(String args[])
 {
 File obj = new File("java/course/july/2023");
System.out.print(obj.getPath());
 }
 }
```

What is the output of the above code?

a. java/course/july/2023
b. java/course/july/
c. java/course/
d. 2023

Correct Answer: a

Detailed Solution:
Test by run.

QUESTION 7:

Which of the following is/are interface(s) in the `java.io` package?

a. FileWriter
b. FileFilter
c. ObjectOutput
d. DataOutput

Correct Answer: b, c, d

Detailed Solution:

FileWriter is a class in the java.io package. All others are interfaces.

QUESTION 8:

Fill in the blanks.

The class DataInputStream extends _____ and implements the interface _____.

a. FileInputStream , FileInput
b. FilterInputStream , DataInput
c. FilterInputStream , FileInput
d. FileInputStream , DataInput

Correct Answer: b

Detailed Solution:

public class DataInputStream extends FilterInputStream implements DataInput.

QUESTION 9:

Which of the following method(s) is not included in InputStream class?

a. available( )
b. reset( )
c. read( )
d. flush( )

Correct Answer: d

Detailed Solution:

flush( ) is not a valid method in the InputStream class**.**

QUESTION 10:

Which of the following package(s) contains a large number of stream classes that provide capabilities for processing all types of data?

a. java.awt.
b. java.io
c. java.net
d. java.util

Correct Answer: b

Detailed Solution:

java.io contains a large number of stream classes that provide capabilities for processing all types of data.

PROGRAMMING IN JAVA

Assignment 8

TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 10 × 1 = 10**

QUESTION 1:

Which of the following method is/are of class button in java.awt package ?

a. getLabel()
b. setLabel(String label)
c. getCurrent()
d. getItem(int index)

Correct Answer: a, b

Detailed Solution:
setLabel(String label) method Changes this button's label to be the String argument and and getLabel() method returns the current label of the button.

QUESTION 2:

Which of the following is **NOT** a class of **java.awt package?**

a. Button
b. Component
c. Dialog
d. Paint

Correct Answer: d

Detailed Solution:
Paint is interface of **java.awt package.**

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QUESTION 3:

Which of the following statement(s) is/are correct?

a. AWT components are platform-independent.
b. AWT follows the MVC (Model View Controller).
c. Swing components are platform-dependent.
d. Swing follows MVC (Model View Controller).

Correct Answer: d

Detailed Solution:
AWT components are platform dependent and not follow MVC.Swing components follow MVC and are platform independent.

QUESTION 4:

When we invoke `repaint()` for a `java.awt.Component` object, the AWT invokes which of the following method ?

a. draw( )
b. show( )
c. update( )
d. paint( )

Correct Answer: c

Detailed Solution:
The repaint() method calls automatically update() method and in turn update() method calls paint( ) method.

QUESTION 5:

Which package provides many event classes and Listener interfaces for event handling?

a. java.awt
b. java.lang
c. java.io
d. java.util

Correct Answer: a

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Detailed Solution:
The class Event is declared in java.awt package which is used for event classes and Listener interfaces for event handling.

QUESTION 6:

Which of the following is TRUE about the following GUI?

a. There is a Frame and two TextFields.
b. There is a Frame with two Labels and two non-editable TextFields.
c. There are two Labels.
d. There is a Frame with two Labels and two TextFields.

Correct Answer: d

Detailed Solution:
There is a Frame with two Labels and two TextFields in the given GUI.

QUESTION 7:

Which of the following is TRUE regarding check box and radio button?

a. Check box is used for single selection item whereas radio button is used for multiple selection.
b. Check box is used for multiple selection items whereas radio button is used for single selection.
c. Both are used for multiple as well as single item selection.
d. Checkbox is always preferred than radio buttons.

Correct Answer: b

Detailed Solution:

Check box is used for multiple selection items whereas radio button is used for single selection. For example, if a form is asking for your favorite hobbies, there might be multiple correct

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answers to it, in that case check box is preferred. And if a form is asking about gender, there

must be only one true option among the multiple choices, in that case radio buttons are used.

QUESTION 8:

Which of the following is TRUE about check box in Java?

a. A check box can be in either an "on" (true) or "off" (false) state.
b. Clicking on a check box changes its state from "on" to "off," or from "off" to "on."
c. A check box can be in an "on" (true) and in "off" (false) state simultaneously. d. Check boxes cannot be grouped together.

Correct Answer: a, b

Detailed Solution:
A check box is a graphical component that can be in either an "on" (true) or "off" (false) state. Clicking on a check box changes its state from "on" to "off," or from "off" to "on.". A check box cannot be in both “on” and “off” state simultaneously. Further, several check boxes can be grouped together under the control of a single object, using the CheckboxGroup class. In a check box group, at most one button can be in the "on" state at any given time. Clicking on a check box to turn it on forces any other check box in the same group that is on into the "off" state.

QUESTION 9:

Which of the following is the latest graphics and media package for Java?

a. Applet
b. AWT
c. Swing
d. JavaFX

Correct Answer: d

Detailed Solution:
JavaFX is a set of latest graphics and media packages in Java that enables developers to design, create, test, debug, and deploy rich client applications that operate consistently across diverse platforms.

More details can be found here: https://docs.oracle.com/javafx/2/overview/jfxpub-overview.htm.

QUESTION 10:

If setText(String text) is a method of Label class, then why it is called non editable?

a. Because user-input is unavailable during runtime.
b. A Label is just a TextField with non-editable property turned on.
c. Labels are editable by default.
d. setText(String text) is not a method of Label class.

Correct Answer: a

Detailed Solution:
A label class is non-editable during runtime by the user, but internal functions can change the text using the setText(String text) function.

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OBJECT ORIENTED PROGRAMMING WITH JAVA

Assignment-9

TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 10× 1 = 10**

QUESTION 1:

Which of the following Swing components inherently support the WindowListener interface?

a. Swing frames (JFrame)
b. Swing checkboxes (JCheckBox)
c. Swing dialogs (JDialog)
d. Swing combo boxes (JComboBox)

Correct Answer: a, c

Detailed Solution:

Swing frames (JFrame) and Swing dialogs (JDialog) Swing components inherently support the WindowListener interface.

QUESTION 2:

Which class in Swing provides a graphical way to display images, icons, or custom graphics?

a. JImage
b. JLabel
c. JImageIcon
d. JDialog

Correct Answer: b

Detailed Solution:

JLabel is used to display images, icons, or custom graphics**.**

QUESTION 3:

Which event listener interface is commonly used to handle user actions like button clicks in Swing?

a. ActionListener
b. MouseListener
c. KeyListener
d. WindowListener

Correct Answer: a

Detailed Solution:

ActionListener interface is commonly used to handle user actions like button clicks in Swing.

QUESTION 4:

What is/are the way(s) to create a Frame in Java Swing?

a. By creating the object of Frame class (association)
b. By extending Frame class (inheritance)
c. By importing a package named JFrame
d. By declaring a class with name JFrame

Correct Answer: a, b

Detailed Solution:

We can create a frame component by creating the object of Frame class (*association*) and by extending Frame class (*inheritance*).

QUESTION 5:

Which method is used to add a component to a JPanel in Java Swing?

a. add ( )
b. addComponent( )
c. addElement( )
d. insertComponent( )

Correct Answer: a

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Detailed Solution:
add( ) - method is used to add a component to a JPanel in Java Swing . Other options are not valid in the context of JPanel.

QUESTION 6:

Which of the following(s) is/are interface(s) of java.awt package?

a. Button
b. Paint
c. Dialog
d. Shape

Correct Answer: b, d

Detailed Solution:
Paint, Shape are interfaces of **java.awt package.**

QUESTION 7:

Which AWT layout manager is often used to create a responsive grid of components that

automatically adjusts to the container's size?

a. BorderLayout
b. GridLayout
c. GridBagLayout
d. BoxLayout

Correct Answer: b

Detailed Solution:

GridLayout manager is often used to create a responsive grid of components that automatically adjusts to the container's size

---

QUESTION 8:

Which Swing component is used for allowing users to select files and directories from the file system?

a. JFileChooser
b. JFileSelector
c. JFilePicker
d. JFileDialog

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Correct Answer: a

Detailed Solution:

QUESTION 9:

What is the primary disadvantage of using AWT over Swing for GUI development?

a. AWT components are not well-suited for complex layouts.
b. AWT lacks support for event handling.
c. AWT components are heavyweight and platform-dependent.
d. AWT does not support graphics rendering.

Correct Answer: c

Detailed Solution:

AWT components are heavyweight and platform-dependent.

QUESTION 10:

Which of the following is/are valid alignment option(s) for components in a FlowLayout?

a. LEFT
b. TOP
c. BOTTOM
d. LEADING

Correct Answer: a, d

Detailed Solution:

The following are valid alignment options for components in a FlowLayout in Java: LEFT, RIGHT, CENTER, LEADING, TRAILING.

**********END**********

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PROGRAMMING IN JAVA

Assignment10

TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 10 × 1 = 10**

PROGRAMMING IN JAVA

Assignment10

TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 10 × 1 = 10**

QUESTION 1:

Which of the following statement(s) is(are) true?

a. TCP is not reliable.
b. UDP is not reliable.
c. TCP is fast and UDP is slow.
d. In HTTPS, all communication between two computers are encrypted.

Correct Answer: b, d

Detailed Solution:

TCP is a reliable but slow. HTTPS is the secure version of HTTP, where all communications between two computers are encrypted.

QUESTION 2:

Which of the following statement(s) is/are true?

a. DatagramSocket is a UDP endpoint API .
b. DatagramSocket is a TCP server API .
c. ServerSocket is a TCP server API.
d. ServerSocket is a TCP client API.

Correct Answer: a,c

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Detailed Solution:

ServerSocket is a TCP server API, and will typically accept connections from client sockets. DatagramSocket is a UDP endpoint API and is used to send and receive datagram packets.

QUESTION 3:

Which of the following is/are interface(s) present in java.net package?

a. InetAddress
b. URLConnection
c. SocketOptions
d. HttpURLConnection

Correct Answer: c

Detailed Solution:
SocketOptions is an interface in java.net package.

QUESTION 4:

Which of the following statement(s) is/are true?

a. URL is the acronym for Uniform Resource Locator.
b. A URL takes the form of a string that describes how to find a resource on the Internet. c. URLs have two main components: the protocol needed to access the resource and the location of the resource.
d. The URL class is present in the java.net package.

Correct Answer: a,b,c,d

Detailed Solution:

All statements are valid.

QUESTION 5:

In the following URL, identify the Resource name?

https://nptel.ac.in

a. https
b. nptel.ac.in
c. ac.in
d. https://nptel.ac.in

Correct Answer: b

Detailed Solution:

Resource name is nptel.ac.in.

QUESTION 6:

Which of the following is/are application layer protocol(s)?

a. TCP
b. UDP
c. HTTP
d. FTP

Correct Answer: c,d

Detailed Solution:

TCP,UDP are transport layer protocols.

QUESTION 7:

In the socket programming, for an IP address, which can be used to find the host name and IP address of a client/ server?

a. The ServerSocket class
b. The Socket class
c. The InetAddress class
d. The Connection interface

Correct Answer: c

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Detailed Solution:
An object of the InetAddress class is used to obtain the IP address and the host name of a client/ server computer connected in a network.

QUESTION 8:

Which of the following statement(s) is/are true?

a. Addresses belongs to a Low-Level API.
b. Sockets belong to High-Level API.
c. URIs and URLs are Low-Level API.
d. Interfaces are High-Level API.

Correct Answer: a

Detailed Solution:

The java.net package can be roughly divided in two sections:

- *A Low Level API*, which deals with the following abstractions:
  - *Addresses*, which are networking identifiers, like IP addresses.
  - *Sockets*, which are basic bidirectional data communication mechanisms.
  - *Interfaces*, which describe network interfaces.
- *A High Level API*, which deals with the following abstractions:
  - *URIs*, which represent Universal Resource Identifiers.
  - *URLs*, which represent Universal Resource Locators.
  - *Connections*, which represents connections to the resource pointed to by *URLs*.

QUESTION 9:

Which of the following statement(s) is/are true?

a. DatagramSocket implements Object.
b. DatagramSocket implements Closeable.
c. DatagramSocket extends Object.
d. DatagramSocket extends Closeable.
e. DatagramSocket extends Object implements Closeable

Correct Answer: e

Detailed Solution:

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public class **DatagramSocket** extends *Object* implements *Closeable* is the class that represents a socket for sending and receiving datagram packets.

QUESTION 10:

Which of the following is/are application layer protocol(s)?

a. TCP
b. UDP
c. ARP
d. SMTP

Correct Answer: d

Detailed Solution:

TCP, UDP are transport layer protocols. ARP is a Network - IP layer protocol. SMTP is Application layer protocol.

**********END**********

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OBJECT ORIENTED PROGRAMMING WITH JAVA

Assignment11

TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 10× 1 = 10**

QUESTION 1:

Which of the following statements is true regarding the executeQuery( ) method of the java.sql.Statement interface in Java Database Connectivity (JDBC)?

a. The executeQuery( ) method is used to execute a DELETE statement.
b. The executeQuery( ) method is used to execute a SELECT statement.
c. The executeQuery( ) method is used to execute an INSERT statement.
d. The executeQuery( ) method is used to execute an UPDATE statement.

Correct Answer: b

Detailed Solution:

The executeQuery( ) method is used to execute a SELECT statement.

QUESTION 2:

In JDBC, which component is responsible for managing a connection to the database?

a. DriverManager
b. Connection
c. Statement
d. ResultSet

Correct Answer: a

Detailed Solution:

DriverManager is responsible for managing a connection to the database.

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QUESTION 3:

What is the primary purpose of a PreparedStatement in JDBC?

a. To execute stored procedures.
b. To execute SQL queries with parameters.
c. To execute batch updates.
d. To retrieve metadata about the database.

Correct Answer: b

Detailed Solution:

The PreparedStatement interface is used to execute parameterized SQL statements in

JDBC. _____

QUESTION 4:

Which exception is commonly thrown when there is a failure in JDBC database connectivity?

a. FileNotFoundException
b. NullPointerException
c. SQLException
d. ClassNotFoundException

Correct Answer: c

Detailed Solution:

SQLException is thrown when there is a failure in JDBC database connectivity.

QUESTION 5:

What is a "JDBC driver" in the context of database connectivity?

a. A database management system.
b. A Java class for executing SQL queries.
c. A software component that provides a bridge between Java applications and a specific database.
d. A tool for generating SQL statements.

Correct Answer: c

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Detailed Solution:

JDBC driver: A software component that provides a bridge between Java applications and a specific database.

QUESTION 6:

Which package in Java contains the classes and interfaces for JDBC?

a. java.sql
b. java.io
c. java.db
d. java.net

Correct Answer: a

Detailed Solution:

java.sql package in Java contains the classes and interfaces for JDBC.

QUESTION 7:

Which method is used to execute a batch of SQL statements in JDBC?

a. executeBatch( )
b. executeUpdateBatch( )
c. executeBatchUpdate( )
d. executeUpdate( )

Correct Answer: a

Detailed Solution:

Batch updates in JDBC allow multiple SQL statements to be executed in a single transaction. They can be used for any type of SQL statement, including INSERT, UPDATE, and DELETE. Batch updates are executed when the batch is executed, either explicitly using the executeBatch( ) method or implicitly when the batch size limit is reached. Batch updates can be rolled back using the rollback( ) method.

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QUESTION 8:

Which method is used to move the cursor to the next row in a ResultSet?

a. nextRow( )
b. next( )
c. moveNext( )
d. moveRow( )

Correct Answer: b

Detailed Solution:

The next() method moves the cursor to the next row.

QUESTION 9:

Which type of JDBC driver is also known as the "Thin driver"?

a. Type 1 driver
b. Type 2 driver
c. Type 3 driver
d. Type 4 driver

Correct Answer: d

Detailed Solution:
JDBC: Type 4 - Thin Driver

The thin driver converts JDBC calls directly into the vendor-specific database protocol. That is why it is known as thin driver. It is fully written in Java language.

QUESTION 10:

Which type of JDBC driver is a pure Java driver that communicates with the database using a middleware server?

a. Type 1 driver
b. Type 2 driver
c. Type 3 driver
d. Type 4 driver

Correct Answer: c

Detailed Solution:

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JDBC: Type 3 - Network Protocol

The network protocol driver uses middleware (application server) that converts JDBC calls directly or indirectly into the vendor-specific database protocol. It is fully written in java.

TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 10× 1 = 10**

QUESTION 1:

Which statement is true in case of using "this" keyword in a static method? a. "this" keyword can be used in static method to access static variables only b. "this" keyword can't be used in static method

c. "this" keyword can be used in static method

d. None

Correct Answer: a

QUESTION 2:

State whether the following statements are True or False.

i) A catch can have comma-separated multiple arguments.

ii) Throwing an Exception always causes program termination.

a. True, False

b. False, True

c. True, True

d. False, False

Correct Answer: d

Detailed Solution:

QUESTION 3:

Which of the following contains both date and time?

a) java.io.date

b) java.sql.date

c) java.util.date

d) java.util.dateTime

Answer: c

Explanation: java.util.date contains both date and time. Whereas, java.sql.date contains only date.

QUESTION 4:

In Graphics class which method is used to draws a rectangle with the specified width and height?

a) public void drawRect(int x, int y, int width, int height)
b) public abstract void fillRect(int x, int y, int width, int height)
c) public abstract void drawLine(int x1, int y1, int x2, int y2)
d) public abstract void drawOval(int x, int y, int width, int height)

ANSWER: a

QUESTION 5:

Which of the following control expressions are valid for an if statement? a. Any integer expression.

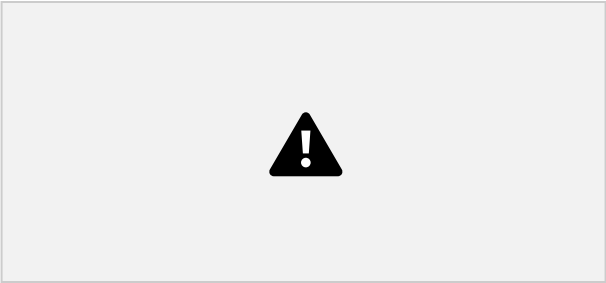
b. Any Boolean expression.
c. A String object.
d. Any expression with mixed arithmetic.

Correct Answer: b

Detailed Solution:
In Java if accepts only boolean expression

QUESTION 6:

Consider the following program:



What is the output of the Java program given above?

A) No output

B) GOAT

C) DOMESTIC

D) Compiler error

Answer D

Explanation:

Case statements should start with either "case" or "default" only.

QUESTION 7:

Consider the following program:

What is the output of the following program?

a. java

b. ava

```
c. y java
d. january
```

Correct Answer: b

Detailed Solution:

QUESTION 8:

Which of the following statements would cause a compilation error?

```
a. float[] = new float(3);
b. float f2[] = new float[];
c. float[] f1 = new float[3];
d. float f3[] = new float[3];
e. float f5[]={ 1.0f, 2.0f, 2.0f };
f. float f4[] = new float[] { 1.0f, 2.0f, 3.0f};
```

Correct Answer: a, b

Detailed Solution:
Option (c), (d), (e) and (f) are syntactically correct for declaration of an array. (a), (b) will cause error.

QUESTION 9:

Consider the following piece of code in Java.

```
public class Test {
 public static void aMethod() throws Exception {
   try {
    throw new Exception();
   }
   finally {
    System.out.print("finally ");
   }
 }
 public static void main(String args[]){
  try {
   aMethod();
  }
  catch (Exception e) {
```

```
    System.out.print("exception ");
    } finally {
     System.out.print("finally ");
    }

   System.out.print("finished ");
  }
}
```

What is the result, if the above-mentioned program is executed?

a. finally

b. finally exception finally finished

c. finally exception finished

d. Compilation fails

Correct Answer: b

Detailed Solution:
The program is syntactically correct and here for two try blocks.

QUESTION 10:

Which is a component in AWT that can contain another components like buttons, textfields, labels etc.?

A) Window

B) Container

C) Panel

D) Frame

Correct Answer: B

Detailed Solution:
The Container is a component in AWT that can contain another components like buttons, textfields, labels etc. The classes that extends Container class are known as container such as Frame, Dialog and Panel.

**********END**********